

EXPERIMENT 19 – Survey Responses Analysis

Question

Analyze survey responses using grouped bar chart, stacked bar chart, and table.

Dataset

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

The supplied Survey Responses dataset and questions are given in Set 15.

R Code

```
# Experiment 19 - Survey Responses
```

```
survey <- data.frame(
```

```
  ID = c(1, 2, 3),
```

```
  Q1 = c("A", "B", "C"),
```

```
  Q2 = c("B", "A", "A"),
```

```
  Q3 = c("C", "D", "B")
```

```
)
```

```
# Question 1 - Grouped Bar Chart
```

```
q1 <- table(survey$Q1)
```

```
barplot(q1,
```

```
  main = "Question 1 Responses",
```

```
  xlab = "Answer",
```

```
  ylab = "Frequency",
```

```
  col = "skyblue")
```

```
# Question 2 - Stacked Bar Chart
```

```

response_matrix <- rbind(
  table(factor(survey$Q1, levels = c("A","B","C","D"))),
  table(factor(survey$Q2, levels = c("A","B","C","D"))),
  table(factor(survey$Q3, levels = c("A","B","C","D")))
)
barplot(response_matrix,
  beside = FALSE,
  main = "Overall Survey Response Distribution",
  xlab = "Question",
  ylab = "Frequency",
  col = c("skyblue", "lightgreen", "orange", "pink"))

```

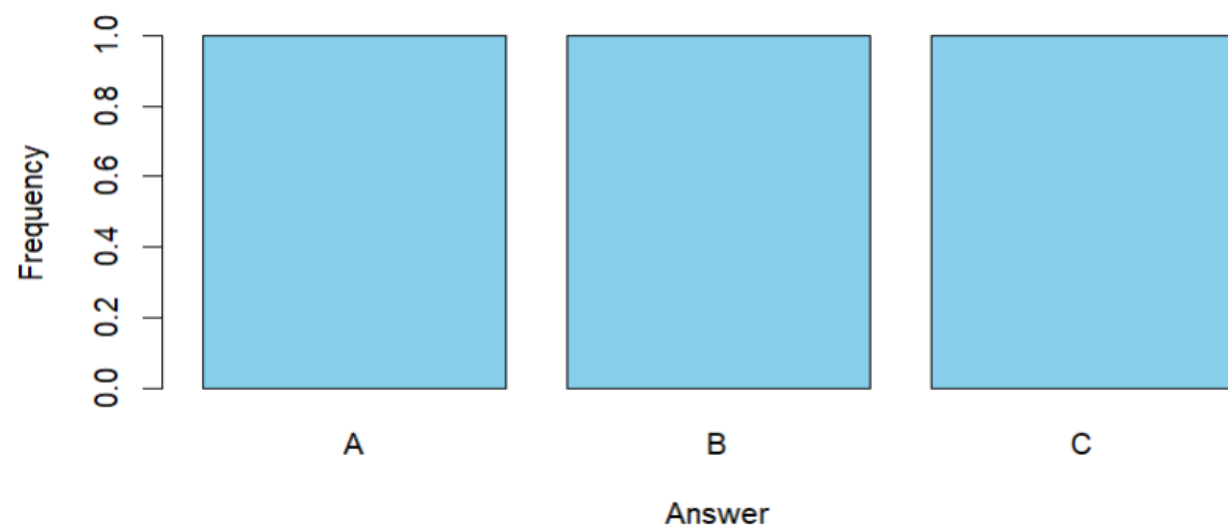
Question 3 - Table

```
print(survey)
```

Question 1

Create a grouped bar chart to visualize the distribution of answers for Question 1.

Output 1

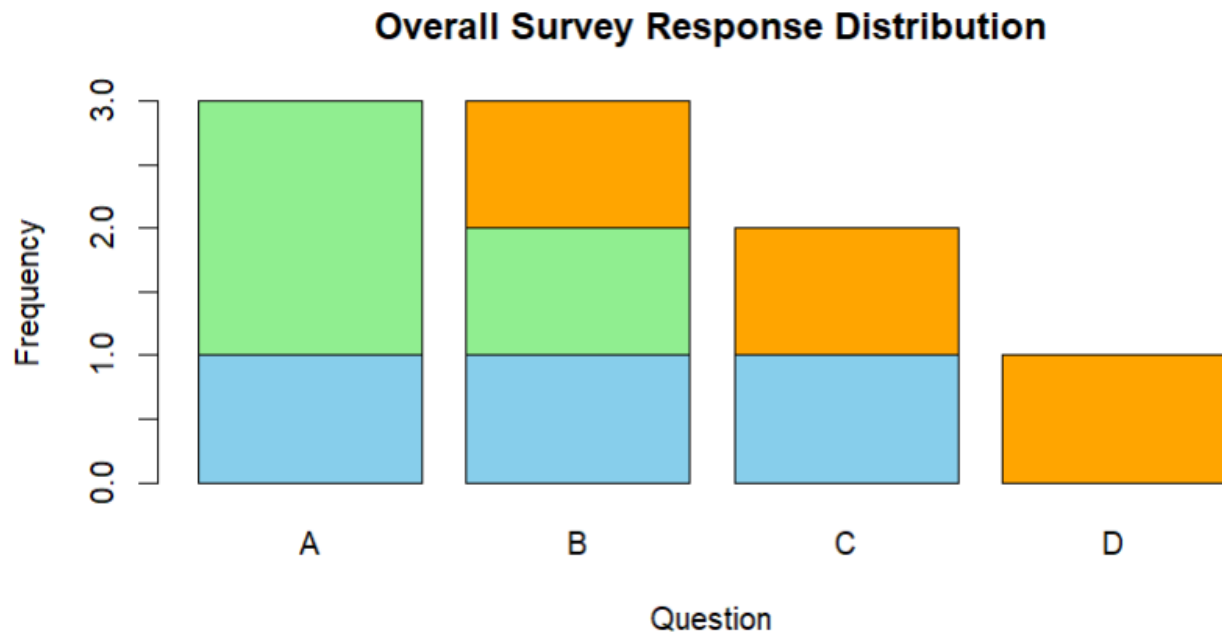


Question 1 contains one response each for A, B, and C.

Question 2

Generate a stacked bar chart showing the overall response distribution.

Output 2



The stacked bar chart compares answer frequencies across Questions 1, 2, and 3.

Question 3

Display survey response data.

Output 3

```
> # Question 3 - Table
> print(survey)
  ID Q1 Q2 Q3
1  1  A  B  C
2  2  B  A  D
3  3  C  A  B
```